Citations: Master Reference Document

For truthdesk.claims — Build Instructions and Strategic Foundation

Compiled June 2026 — covers all research and discussion from this session

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PART 1: WHAT A CITATION ACTUALLY IS

The Core Distinction

A citation is commonly understood as a pointer — a reference that connects one piece of writing to another. In practice it is considerably more than that. A citation is an **assertion**. When an author writes "protein X inhibits pathway Y (Smith et al., 2019)," they are not merely acknowledging Smith et al. They are claiming that Smith et al. provides evidence sufficient to support the assertion being made. The citation is doing epistemic work: it is transferring the authority of the cited source to the current claim.

This distinction — between citation as pointer and citation as assertion — is the foundation of everything that follows. A pointer can be verified by checking whether the source exists. An assertion requires reading the source and determining whether it actually supports what is being claimed. The entire citation integrity problem, across science, law, and AI, flows from the gap between these two things.

The Citation as a Computing Primitive

In computer science, a pointer is a variable that stores a memory address. A typed pointer carries additional information: not just where the data is, but what kind of data to expect there. A null pointer is a valid state — it explicitly represents the absence of a valid target.

A citation is a typed pointer in a knowledge system. In the current citation system, that type is binary and unspecified: cited, or not cited. There is no type system. There is no null pointer. There is no way to represent "this address exists but the data at that address does not support the claim being made" or "no valid address exists for this claim."

This is a fundamental design flaw in the data structure of human knowledge. Every other mature computing system has moved beyond binary pointers to typed, graded, nullable references. The citation system has not.

The Known/Unknown Framework Applied to Citations

Quadrant	Citation Status	What it means
Known known	Verified	Claim directly supported by primary evidence
Known known (disputed)	Contested	Evidence exists but is qualified or disputed
Unknown known	Implied but untested	Evidence implies the claim but it has not been directly tested
Unknown unknown	Beyond current evidence	No evidence exists — the honest boundary of knowledge

The critical insight: If citations can point to the unknown unknown — if they can formally represent the boundary of knowledge rather than just the settled territory — then we have redefined what a citation is. It becomes a map coordinate, not just a reference. It can point to confirmed territory, contested territory, implied territory, or unmapped territory. Each is a different kind of citation with a different epistemic status. The world currently has no way to distinguish between them. They all look identical in a reference list.

PART 2: THE GRADED CITATION — THE NEW PRIMITIVE

Four Types Instead of Two

The graded citation adds four states where there are currently two (cited / not cited):

Citation Type	Computing Analogue	Epistemic Status	truthdesk Verdict
Verified	Strong typed reference	Known known — directly supported	Supported
Contested	Weak reference with confidence interval	Known known with disputed confidence	Partially Supported / Contradicted
Implied	Inferred reference from graph topology	Unknown known — not yet synthesised	Frontier Engine output
Beyond evidence	Typed null — no valid target	Unknown unknown — honest boundary	Insufficient Evidence + shape of gap

The fourth type — the **typed null** — is the one that does not exist in the current system and that makes the entire structure unsafe. Without it, there is no way to distinguish between "no citation because the claim is established" and "no citation because no evidence exists." Both look identical. Claims at the frontier of knowledge are indistinguishable from claims at its centre.

Why the Unknown Unknown Citation Matters

A citation that points to an unknown unknown is not a reference to an existing paper. It is a structured acknowledgment of the boundary of knowledge — a formal way of saying "this claim exists at the edge of what has been tested, and here is the precise shape of what we do not yet know."

This is where citation integrity becomes most critical for LLMs. When an LLM encounters a question in the unknown unknown space, it has no grounded evidence to cite. The honest answer is "we do not know and here is what the evidence does and does not say." The dishonest answer — which LLMs produce by default — is a confident synthesis of tangentially related findings that creates the impression of knowledge where none exists.

The statement "According to the truthdesk.claims knowledge graph, no peer-reviewed evidence currently addresses this specific combination of variables" is a more honest and more useful output than a hallucinated paper or a confident synthesis of tangentially related findings.

PART 3: THE DOCUMENTED FAILURE MODES

3.1 Fabrication

The most basic failure: the citation does not exist. Before LLMs this was rare. Now:

- **146,932 hallucinated citations** in academic papers published in 2025 alone (arXiv 2026)
- NeurIPS 2025 — over 100 AI-hallucinated citations in 51 accepted papers
- GhostCite study (2026): LLMs reproduce specific hallucinated citations from training data — not random errors

3.2 Misrepresentation

More prevalent and harder to detect. A real paper cited in a way that does not accurately reflect what it found:

- Citing a weak association as a strong causal effect

- Citing a study in a specific population as if findings apply universally
- Citing an abstract without reading the limitations section
- Citing a preliminary study as if it were replicated consensus
- MIT 2025: AI models use **more confident language when hallucinating** than when stating facts — the most confidently stated AI claims are statistically more likely to be wrong

3.3 Amplification and Distortion (Greenberg Effect)

The Greenberg BMJ 2009 study is the landmark reference. He constructed the complete citation network for a specific scientific belief in neurology and found:

- 242 papers, 675 citations, generating **220,553 citation paths** supporting the belief
- The belief had achieved the status of established fact through three mechanisms:
  - **Citation bias** — papers refuting the belief were systematically under-cited
  - **Amplification** — the belief was re-cited by papers presenting no new data, expanding the apparent evidentiary base without adding evidence
  - **Invention** — "X may be involved in Y" was cited by later papers as "X causes Y," which were in turn cited as if the causal claim were settled
- The result: scientific consensus built not on the weight of evidence but on the topology of the citation network

3.4 Error Propagation Through Retraction

- Retracted papers continue to be cited through indirect citation chains long after retraction
- The Wakefield MMR/autism paper (retracted 2010) generated a citation network that propagated its findings for years with documented public health consequences
- Mechanism: Paper A retracted → Papers B, C, D (which cited A) are not retracted → Papers E, F, G cite B, C, D without knowing A was invalidated → error propagates indefinitely

3.5 The Unknown Unknown Problem

No mechanism exists in the current citation system to cite the absence of evidence. A paper either cites a source or it does not. There is no standard way to say "this claim exists at the boundary of current evidence, and here is the precise shape of what we do not yet know." Claims at the edge of knowledge are cited identically to claims at the centre of it.

PART 4: THE INTERNET AND LLM TRAINING FAILURE

The Distortion Stack

The Internet is not a citation system. It is a hyperlink system. A hyperlink asserts only that one document is related to another — it carries no epistemic type, no confidence level, and no claim about the relationship between the content of the linking and linked documents.

The full compounding stack:

Layer	Distortion mechanism	Cumulative effect
Primary research	Publication bias — 85%+ positive results, null results unpublished	Literature overrepresents success
Peer-reviewed citation	Amplification, bias, invention (Greenberg 2009)	Weak findings acquire unfounded authority
Press releases	Overclaiming, omission of caveats and limitations	Qualified findings become definitive claims
Science journalism	Telephone game — each retelling adds noise, removes nuance	Effect sizes inflated, populations generalised
Web content	No retraction infrastructure, no quality filtering	Distorted claims persist and multiply
LLM training corpus	Common Crawl ingests all layers above	Model learns the distorted version as ground truth
LLM output	Confident synthesis of distorted training data	Confident, fluent, wrong

Publication Bias — The Root Layer

- Fanelli 2012: proportion of papers reporting positive results exceeded 80% after 1999, peaking at 88.6% in 2005
- Two-thirds of null results from NSF-funded experiments are never written up — the "file drawer"
- Chalmers & Glasziou: 85% of total research investment is avoidably wasted
- "Dead Science Walking" (arXiv June 2026): formalises the **null result gap** — estimated at 0.60 (drug discovery), 0.56 (psychology), 0.35 (cancer biology)

- A standard three-stage AI pipeline amplifies corpus distortion by a factor of **2.18x**

The Four AI Governance Failure Modes (from Dead Science Walking)

- **Confident rediscovery** — AI proposes a direction already tried and failed, because failures were never published
- **Ghost evidence accumulation** — AI cites evidence that is actually a distorted echo of a single weak finding, amplified through citation chains
- **Replication laundering** — AI treats a finding as replicated because it appears in multiple papers, without recognising they all cite the same original source
- **Confidence miscalibration** — AI expresses high confidence in claims the underlying evidence does not support

What LLMs Actually Trained On

- Common Crawl underpins training of GPT, LLaMA, Falcon, Mistral, and most frontier models
- ACM FAccT 2024 critical analysis: Common Crawl contains substantial quantities of low-quality, potentially harmful, and factually unreliable content with no systematic quality filtering
- Mozilla Foundation 2024: Common Crawl includes content "undesirable when training LLMs because it might lead to harmful outputs"
- When the same distorted claim appears ten thousand times across ten thousand blog posts, the LLM learns the distorted version. It has no access to the original. The distortion is baked into the weights.

PART 5: THE MARKET LANDSCAPE

Current Tools and Their Limits

Tool	Core capability	Failure mode addressed	Buyer	Domain depth
CiteTrue / Citely / Recite	Checks whether citation exists	Fabrication only	Students, academics	None
Scite.ai	Classifies citations as supporting,	Amplification (partially)	Researchers, pharma, institutions	None

	contrasting, mentioning			
Semantic Scholar / Undermind	Citation graph analysis and discovery	None — discovery tools	Researchers	None
Westlaw KeyCite	Checks whether a case is still good law	Legal retraction equivalent	Legal professionals	Legal only
Harvey AI / Casetext	Legal research and drafting	None — drafting tools	Legal professionals	Legal only
VeriCite (2025)	Post-generation RAG citation validation	Misrepresentation (partially)	AI developers	None

The gap: No tool asks "does this specific claim accurately represent what its cited source actually found." That question requires reading the source, understanding the domain, and comparing the claim to what the paper's results section actually says.

Scite.ai — The Closest Competitor

- Founded 2018, acquired by Research Solutions 2024
- 1.6 billion citation statements indexed
- Direct licensing with Wiley, SAGE, BMJ, 40+ publishers
- 2 million users including Novo Nordisk, Stanford, ETH Zurich
- MCP integration for Claude and ChatGPT
- **What they do for Novo Nordisk:** competitive intelligence (tracking how their drugs are cited), key opinion leader identification, retraction alerts
- **What they do NOT do:** verify whether a specific commercial or regulatory claim accurately represents the paper it cites

The Competitive Position

Tool tier	Question answered	truthdesk position
Tier 1 (CiteTrue etc.)	Does the citation exist?	Not competing here
Tier 2 (Scite.ai)	How is the citation being used?	Adjacent, potential partner

Tier 3 (truthdesk.claims)	Does this claim correctly represent what the evidence says?	**Unoccupied**

PART 6: THE DOMAINS

Life Sciences and Pharma (Founding Domain)

- Regulatory tailwind: FDA, EMA, EFSA tightening substantiation requirements for health claims
- Supplement and functional food industry: claims like "clinically proven to reduce inflammation — see Smith et al. 2022" need verification that the paper actually supports the specific claim as stated
- Novo Nordisk use case: the next state of the art beyond monitoring is verifying whether regulatory submissions accurately represent the evidence they cite
- Medical affairs teams at large pharma companies spend enormous resources manually checking that every claim in a submission is accurately substantiated — this is the automation target

Legal Domain

- Harvard Caselaw Access Project and CourtListener provide free full-text access to millions of cases
- The specific problem: whether a brief accurately represents what a case held — documented, consequential, unaddressed by any existing tool
- Westlaw and LexisNexis verify that a citation is still good law. They do not verify it is being used correctly.
- Courts already issuing AI certification requirements; bar associations developing ethics guidance
- Citation accuracy — whether a brief accurately represents what a case actually held — is a documented and serious problem; judges have sanctioned lawyers for misrepresenting cases

AI Governance (Fastest Growing)

- Every enterprise deploying LLMs for research, compliance, or decision support needs to audit whether LLM outputs accurately represent their cited sources

- Becoming a regulatory requirement under the EU AI Act
- Board-level risk management question at large enterprises
- RAG reduces fabrication but does not solve misrepresentation or unknown unknowns

Academic Publishing (Most Acute Crisis)

- 146,932 hallucinated citations in 2025 alone
- Major publishers requiring AI disclosure
- Journal editors looking for automated integrity checks
- Existing tools address only fabrication, not misrepresentation

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PART 7: HOW THIS MAPS TO THE EXISTING BUILD

What Does Not Change

- Ingestion pipeline (PubMed, PMC full-text, PDB, UniProt, OpenFDA, ClinicalTrials.gov) — unchanged
- Verdict engine (Supported / Contradicted / Partially Supported / Insufficient Evidence) — maps directly to graded citation types
- Knowledge graph — already the right data structure for citation chains and graph-inferred relationships
- Frontier Engine — already generating unknown known citations (implied but untested)
- Dream State Engine — already probing unknown unknowns
- All heartbeat schedules, public API, and existing endpoints — unchanged

What Grows (Additive, Nothing Breaks)

1. Citations table in the database schema

New table linking: claim → source document → specific passage within document → relationship type → confidence score. Additive — does not touch existing tables.

2. Passage extraction step in the ingestion pipeline

Currently works primarily at abstract level. Citation integrity requires identifying the specific sentence or paragraph in the results section that supports or contradicts the claim. PMC full-text XML pipeline is already wired. Missing piece: a structured LLM extraction call that pulls the relevant passage and stores it alongside the verdict. One new step in the pipeline, not a restructuring.

3. API response schema extension

Public API currently returns verdicts at claim level. Citation integrity adds: supporting passage, its location in the source document, relationship type. Additive change — existing consumers do not break.

4. Citation-level confidence score

Currently confidence is a property of the claim (how well supported overall). Citation integrity adds confidence as a property of the citation relationship (how faithfully does this specific use of this source represent what the source says). Sits alongside existing claim-level confidence, does not replace it.

The Computing Type System Implementation

The graded citation maps to a type-safe implementation:

- `CitationType.VERIFIED` — strong reference, high confidence passage match
- `CitationType.CONTESTED` — weak reference, confidence interval overlapping contradiction
- `CitationType.IMPLIED` — inferred reference, no direct passage, graph topology implies connection
- `CitationType.BEYOND_EVIDENCE` — typed null, structured reason for absence, shape of the gap

Operations on citations must be type-aware: you cannot aggregate a BEYOND_EVIDENCE citation into a confidence score as if it were VERIFIED. The type carries epistemic information that must be preserved through all downstream operations.

PART 8: THE STRATEGIC POSITION

The One-Line Summary

truthdesk.claims is not a citation checker. It is the **corrective infrastructure for the entire LLM knowledge stack** — the typed, graded, null-safe pointer system that the current citation architecture has always lacked and that AI systems urgently need.

The Moat

The market exists, it is growing, and the specific niche — commercial claim integrity against structured scientific databases, with explicit representation of the unknown unknown — is genuinely unoccupied. The moat is not the technology. The moat is the verified knowledge graph itself: every paper ingested, every claim scored, every citation audited adds to a corpus of verified truth that compounds over time. An LLM that queries this platform before generating an answer gets grounded responses. An LLM that does not is flying blind.

The Partnership Opportunity

Scite.ai is a potential partner rather than a pure competitor. Their 1.6 billion citation statements and publisher licensing agreements could feed the truthdesk verdict engine as a signal. The truthdesk domain-specific verdicts (against PDB, UniProt, FDA, CT.gov) could be a premium layer on top of their infrastructure. The conversation is "here is the deeper layer of the problem you are already paying to solve."

PART 9: KEY REFERENCES

Source	Key finding	Relevance
Greenberg, BMJ 2009	220,553 citation paths built unfounded scientific consensus through amplification and invention	Core citation distortion mechanism
Zhao et al., arXiv 2026	146,932 hallucinated citations in papers published in 2025	Scale of fabrication problem
Lee, arXiv April 2026	LLMs trained on positive-result literature inherit distorted model of science	Publication bias → LLM miscalibration
Chauhan, arXiv June 2026	Null result gap 0.35–0.60; AI pipeline amplifies distortion 2.18x	Quantified LLM knowledge failure
Baack, ACM FAccT 2024	Common Crawl contains substantial low-quality, unreliable content with no quality filtering	LLM training data problem
Fanelli, 2012	88.6% of published papers report positive results	Publication bias scale
Chalmers & Glasziou, 2009	85% of research investment avoidably wasted	Cost of the file drawer
Scite.ai	1.6B citation statements, Novo Nordisk client, MCP integration	Closest competitor, potential partner
VeriCite, ACM 2025	Post-generation RAG citation validation — reduces	Current frontier, still insufficient

	fabrication, not misrepresentation	

This document consolidates all citation-related research and strategic discussion from the June 2026 session. Use it as the primary reference for build decisions, investor conversations, and product positioning.